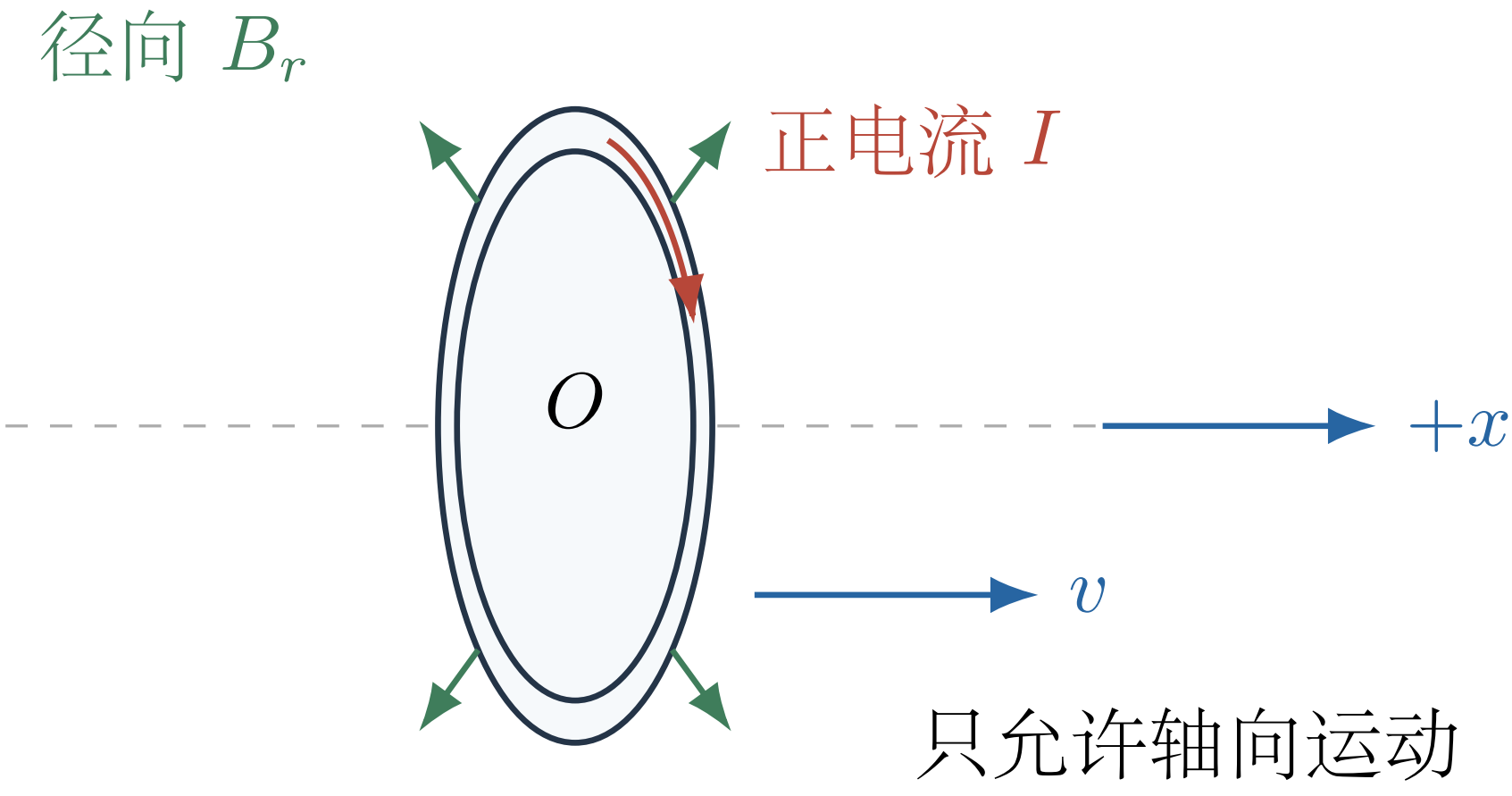
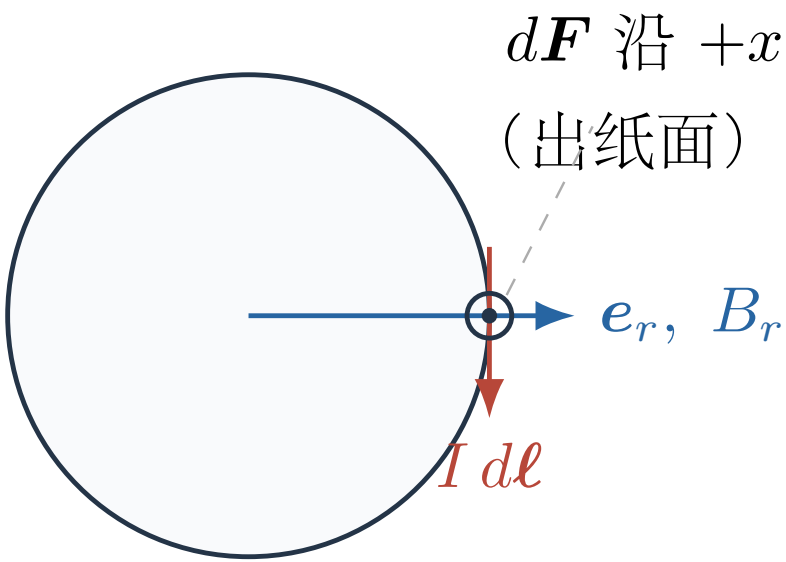


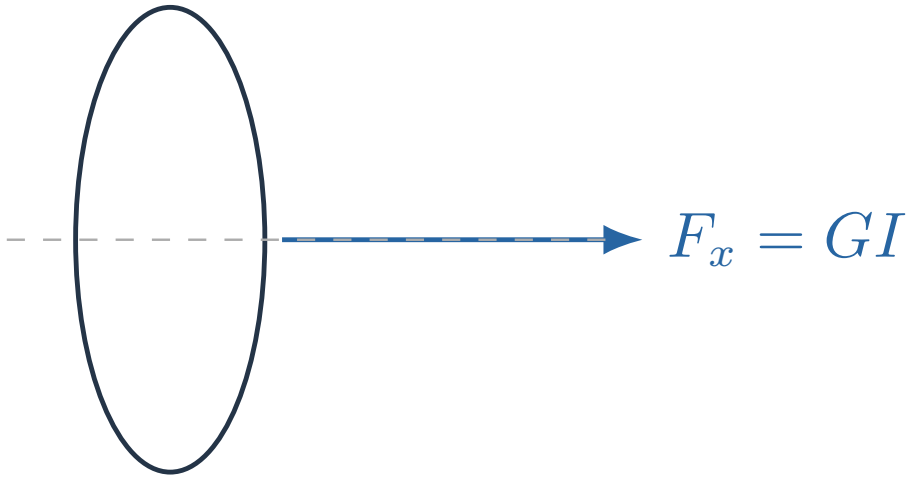
n 匝线圈，半径 R

线圈平面垂直于 x 轴

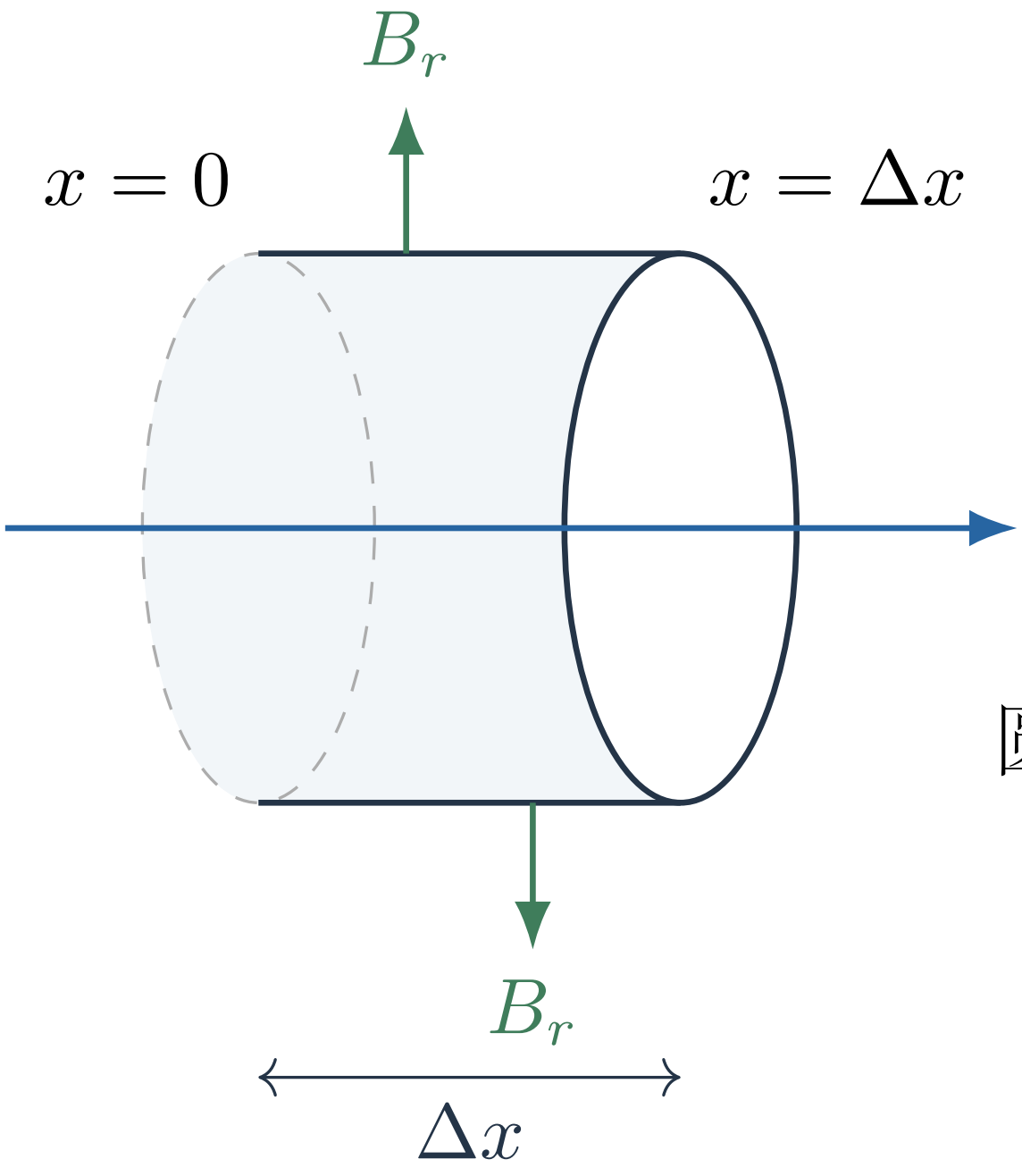


横截面观察

正电流沿顺时针



整体轴向受力

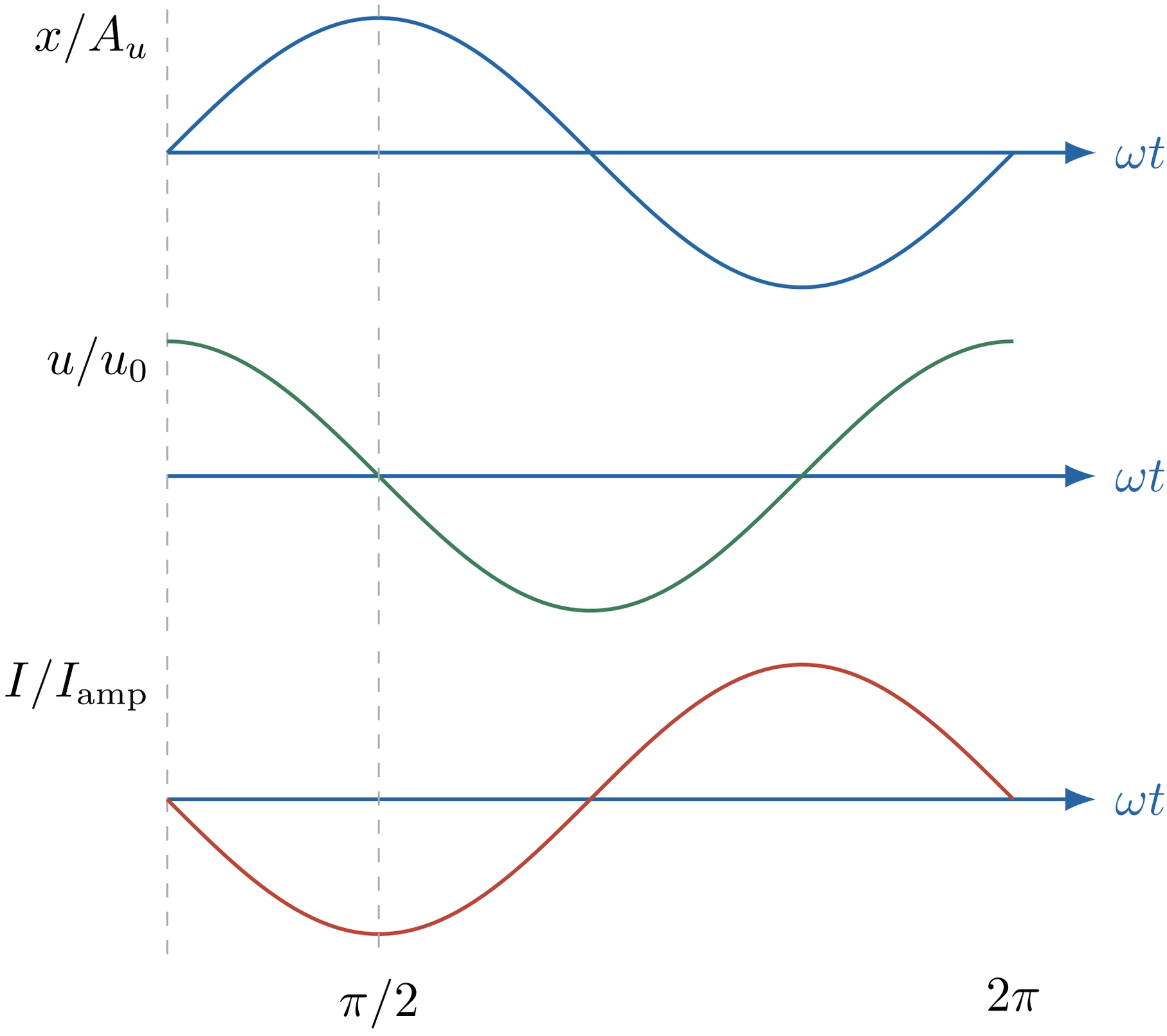


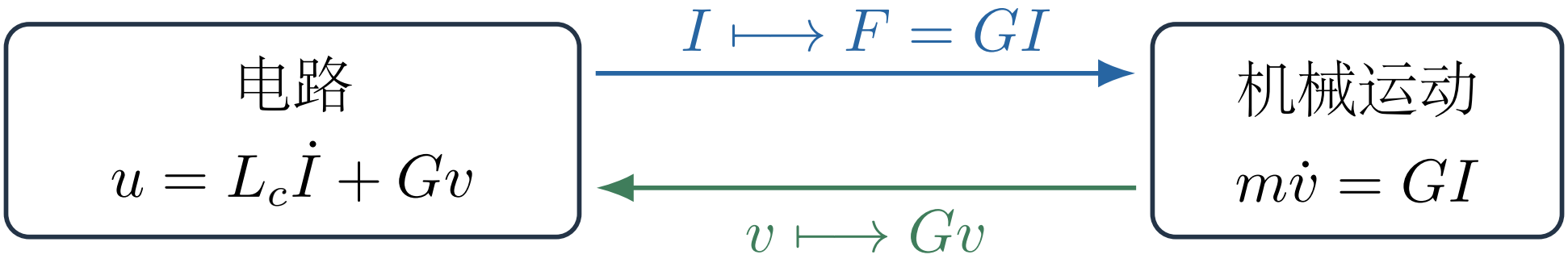
侧面积 $2\pi R\Delta x$

$\Delta\Psi_{\text{ext}} = G\Delta x$

圆面正法向取 $-x$

电压驱动，$L_c = 0$，向右过零取 $t = 0$





$$uI = \frac{d}{dt}\left(\frac{1}{2}mv^2 + \frac{1}{2}L_cI^2\right)$$